Flux Linkages of One Conductor in a Group of Conductors

Consider a group of parallel conductors 1, 2, 3..... *n* carrying currents I_1, I_2, I_3 ..... I_n respectively, as illustrated in Fig. Let it be assumed that the sum of the currents in various conductors is zero *i.e.*, $I_1 + I_2 + I_3 + ... I_n = 0$.

